DANIEL KARIMI

SYSTEMS & QA ENGINEER · VISION & BIOMETRICS SPECIALIST

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PROFESSIONAL SUMMARY

Telecommunications Systems Engineer (Universidad Politécnica de Madrid) with practical expertise in Validation and Verification (V&V) processes, automated testing, and isolating complex issues in End-to-End (E2E) architectures.

At the Spanish National Research Council (CSIC), I engineered testing environments to validate deep learning models (YOLOv8) handling complex image, sensor, and tracking datasets—a methodology highly analogous to testing and vindicating **multi-factor biometric systems**. I excel at writing automated scripts (Python/Linux) to securely test algorithmic precision and data normalization.

I am highly motivated to leverage my rigorous background in systemic troubleshooting and software/hardware integration to bridge the gap between development and final deployment, ensuring resilient digital security solutions.

CORE COMPETENCIES

🧪 QUALITY ASSURANCE (V&V)

E2E Validation & Integration •
Test Automation (Python) •
Test Environment Config •
Isolating & Reproducing Issues •
Performance Analytics

👁️ BIOMETRICS & VISION

Image Processing (OpenCV) •
Target Characterization •
Object Detection (YOLO) •
Data Normalization •
Algorithmic Validation

💻 SOFTWARE DEVELOPMENT

Python / C++ / Bash •
Version Control (Git) • Test Scripting •
Linux / Docker Environments •
Agile Methodologies

⚙️ SYSTEM INTEGRATION

Hardware & Software Convergence •
Backend System Knowledge •
Requirements Analysis •
Technical Documentation •
Troubleshooting

Languages: Spanish (Native) • English (B2 – Aptis ESOL, British Council) — Fluent in creating testing methodologies and cross-team communication.

PROFESSIONAL EXPERIENCE

Systems Integration & Validation Intern (Computer Vision/Sensors) | CSIC – Institute of Physical and Information Technologies (ITEFI)

Sep 2025 – Feb 2026

- Developed the validation strategy and automated test cases to evaluate the precision of YOLOv8 object detection pipelines on complex tracking datasets, analogous to biometric matching protocols.
- Maintained Linux-based validation environments, writing extensive Python scripting to ingest raw sensor data, normalize it, and isolate predictive anomalies.
- Cooperated directly with cross-disciplinary researchers to reproduce, identify, and isolate algorithmic issues, iteratively improving E2E pipeline accuracy by over 15%.
- Authored comprehensive project administration documents, user-guides, and test execution reports detailing the algorithmic trade-offs based on the product's core requirements.
- Applied stringent Validation and Verification (V&V) practices on data interfaces, simulating real-world embedded system hardware constraints against software performance metrics.

E2E Validation | Test Automation | Python/Linux | Biometrics Proxy (Vision/YOLO) | Troubleshooting

EDUCATION

Telecommunications Systems Engineering | **Universidad Politécnica de Madrid (ETSIT-UPM)** *Sep 2020 – Jul 2025*

- **Systems Architecture:** Deep understanding of the convergence between hardware components (sensors, endpoints) and backend processing systems, TCP/IP, and OSI models.
- **Programming & Integration:** Strong foundations in OOP (C, C++, Python, Java), enabling rapid understanding of complex system codebases to develop effective test plans.
- **Systems Engineering & V&V:** Extensive academic background in Validation and Verification methodologies, ensuring software complies with stringent telecommunication standards.
- **Digital Signal Processing (DSP):** Practical usage of filtering and data normalization, crucial for both telecommunications hardware and biometric image pre-processing.

Systems Integration | V&V Methodologies | Backend Knowledge | Hardware/Software Convergence

Technology Baccalaureate, **IES Isabel la Católica** *2018 – 2020*

Science and Technology track. Honours. Rigorous foundation in mathematics and physics.

HIGHLIGHTED ACADEMIC & ENGINEERING PROJECTS

Scalable Hardware-Software Validation Pipeline AUTOMATED TESTING

PYTHON • LINUX • ALGORITHM TESTING

Designed a robust architecture to ingest and validate massive unstructured datasets.

Developed a scalable data-handling pipeline capable of curating and filtering data, passing cleanly structured metrics to the next subsystem. The core focus was to assert execution stability over disparate environments.

- Wrote Python scripts acting as automated test agents to verify downstream algorithm accuracy and isolate structural bottlenecks.
- Engineered reproducible systems demonstrating the ability to bridge hardware-level physical constraints with high-level software deployments effectively.